



Certificate of Calibration

Certificate No. : WK2504-402-59

Page: 1 of 2

Customer : บริษัท เออาร์พี เทคโนโลยี จำกัด (สำนักงานใหญ่)
329/140 หมู่ที่ 4 ตำบลบ้านคลอง
อำเภอเมืองพิษณุโลก จังหวัดพิษณุโลก 65000

Instrument : Earth Tester
Manufacturer : Sanwa
Model : PDR4000
Serial No : 18075302397
Identity No : N/A
Range : See to data
Resolution : See to data

Ambient Temperature : $(23 \pm 2) ^\circ\text{C}$
Humidity : $(50 \pm 15) \% \text{RH}$
Received Date : 02-Apr-25
Calibrated Date : 02-Apr-25
Issued Date : 02-Apr-25
Calibration Location : In Lab

Calibration Method : CP-WK-E01

Reference Standard Instrument :

<u>Instrument</u>	<u>Serial No</u>	<u>Certificate No</u>	<u>Due Date</u>	<u>Traceability to</u>
Multi-Product Calibrator	752021	6011-KL-L1030-24	19-Sep-26	CMI
Resistance Box	3063G19	WK2503-402-232	3-Mar-26	WK Electric Co.,Ltd.

CMI : Czech Metrology Institute

This result calibrate was found accurate as shown on date place of calibrate only

This certificate is traceability to the International System of Unit (SI).

The reported expanded uncertainty it was multipiled by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Mr. Kritsada Oupparattha

Approved by :

Mr. Ratchadawut Bungravee
Authorized Signatory

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Calibration Results

Certificate No.: WK2504-402-59

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Calibration Results

AC Voltage Calibration @ Test Frequency : 50 Hz

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
400 V	100 V	100.2 V	0.10 V	97.7 ~ 102.3 V
	200 V	200.3 V	0.10 V	195.7 ~ 204.3 V
	380 V	380.4 V	0.20 V	372.1 ~ 387.9 V

Resistance Calibration

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
40 Ω	10 Ω	10.04 Ω	0.010 Ω	8.00 ~ 12.00 Ω
	20 Ω	20.01 Ω	0.010 Ω	18.80 ~ 21.20 Ω
	38 Ω	37.96 Ω	0.010 Ω	35.72 ~ 40.28 Ω
400 Ω	100 Ω	100.4 Ω	0.10 Ω	97.7 ~ 102.3 Ω
	200 Ω	200.5 Ω	0.10 Ω	195.7 ~ 204.3 Ω
	380 Ω	380.1 Ω	0.10 Ω	372.1 ~ 387.9 Ω
4000 Ω	1000 Ω	999 Ω	1.0 Ω	977 ~ 1023 Ω
	2000 Ω	2019 Ω	1.0 Ω	1957 ~ 2043 Ω
	3800 Ω	3825 Ω	1.0 Ω	3721 ~ 3879 Ω

(X) Without Adjustment () After Adjustment

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**** End of Certificate****



Certificate of Calibration

Certificate No.: WK2504-402-58

Page : 1 of 3

Customer : บริษัท เออาร์พี เทเลคอม จำกัด (สำนักงานใหญ่)
329/140 หมู่ที่ 4 ตำบลบ้านคลอง
อำเภอเมืองพิษณุโลก จังหวัดพิษณุโลก 65000

Instrument : Insulation Tester
Manufacturer : Uni-T
Model : UT 511
Serial No. : N/A
Identity No. : N/A
Range : See to data
Resolution : See to data

Ambient Temperature : $(23 \pm 2) ^\circ\text{C}$
Humidity : $(50 \pm 15) \% \text{RH}$
Received Date : 02-Apr-25
Calibrated Date : 02-Apr-25
Issued Date : 02-Apr-25
Calibrated Location : In Lab

Calibration Method : Manufacturer's Manual

Standard Instrument

<u>Instrument</u>	<u>Serial No.</u>	<u>Certificate No.</u>	<u>Due Date</u>	<u>Traceability to</u>
Multi-Product Calibrator	752021	6011-KL-L1030-24	19-Sep-26	CMI
High Voltage Digitalmeter	3740226	E1U2501375	4-Apr-26	NA
Resistance Insulation Box	02092019	WK2503-402-233	31-Mar-26	WK Electric Co., Ltd.
Resistance Box	3063G19	WK2503-402-232	3-Mar-26	WK Electric Co., Ltd.

CMI : Czech Metrology Institute

NA : Caltechnologies Co., Ltd.

This result calibrate was found accurate as shown on date place of calibrate only

This certificate is traceability to the International System of Unit (SI)

The reported expanded uncertainty it was multipiled by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Mr. Kritsada Oupparattha

Approved by :

Mr. Ratchadawut Rungravee
Authorized Signatory

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Calibration Results

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Calibration Results

DC Voltage Calibration

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
1000 V	100 V	101 V	1.0 V	95 ~ 105 V
	300 V	301 V	1.0 V	291 ~ 309 V
	500 V	502 V	1.0 V	487 ~ 513 V
	700 V	702 V	1.0 V	683 ~ 717 V
	900 V	902 V	1.0 V	879 ~ 921 V

AC Voltage Calibration @ Frequency : 50 Hz

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
750 V	100 V	98 V	1.0 V	93 ~ 107 V
	300 V	300 V	1.0 V	291 ~ 309 V
	500 V	502 V	1.0 V	487 ~ 513 V
	700 V	704 V	1.0 V	683 ~ 717 V

Insulation Calibration

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
<u>Test Voltage : 100 V</u>				
0.1 to 500 M Ω	1 M Ω	1.0 M Ω	2.9 k Ω	0.5 ~ 1.5 M Ω
	10 M Ω	10.3 M Ω	29 k Ω	9.2 ~ 10.8 M Ω
	100 M Ω	99.8 M Ω	0.29 k Ω	96.5 ~ 103.5 M Ω
<u>Test Voltage : 250 V</u>				
0.5 M Ω to 2 G Ω	1 M Ω	1.0 M Ω	2.9 k Ω	0.5 ~ 1.5 M Ω
	10 M Ω	10.0 M Ω	29 k Ω	9.2 ~ 10.8 M Ω
	100 M Ω	99.8 M Ω	0.29 k Ω	96.5 ~ 103.5 M Ω
	1 G Ω	0.99 G Ω	5.8 M Ω	0.90 ~ 1.10 G Ω
<u>Test Voltage : 500 V</u>				
1 M Ω to 4 G Ω	1 M Ω	0.9 M Ω	2.9 k Ω	0.5 ~ 1.5 M Ω
	10 M Ω	9.9 M Ω	29 k Ω	9.2 ~ 10.8 M Ω
	100 M Ω	99.1 M Ω	0.29 k Ω	96.5 ~ 103.5 M Ω
	1 G Ω	0.98 G Ω	5.8 M Ω	0.90 ~ 1.10 G Ω
<u>Test Voltage : 1000 V</u>				
2 M Ω to 10 G Ω	10 M Ω	9.9 M Ω	29 k Ω	9.2 ~ 10.8 M Ω
	100 M Ω	98.9 M Ω	0.29 k Ω	96.5 ~ 103.5 M Ω
	1 G Ω	0.98 G Ω	5.8 M Ω	0.90 ~ 1.10 G Ω
	10 G Ω	9.7 G Ω	58 M Ω	9.45 ~ 10.55 G Ω

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Calibration Results

Certificate No.: WK2504-402-58

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Calibration Results (Cont.)

DC Output Voltage Calibration

UUC Range	UUC Setting	Measured Value	Uncertainty ($\pm$)	Tolerance Limit Values
100 V	100 V	105 V	1.2 V	100 ~ 120 V
250 V	250 V	255 V	1.2 V	250 ~ 300 V
500 V	500 V	511 V	1.2 V	500 ~ 600 V
1000 V	1000 V	1042 V	1.2 V	1000 ~ 1200 V

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**** End of Certificate****



Certificate of Calibration

Certificate No.: WK2504-402-60

Page : 1 of 3

Customer : ZIRCOZ CO.,LTD.
29/1293 Mooban Wararak Rangsit Khlongsam, Moo.2, Khlong sam,
Khlong Luang , Pathumthani 12120

Instrument : Digital Clamp Meter
Manufacturer : Kyoritsu
Model : KEW 2117R
Serial No. : 1503170
Identity No. : N/A
Range : See to data
Resolution : See to data

Ambient Temperature : $(23 \pm 2) ^\circ\text{C}$
Humidity : $(50 \pm 15) \% \text{RH}$
Received Date : 02-Apr-25
Calibrated Date : 02-Apr-25
Issued Date : 02-Apr-25
Calibrated Location : In Lab

Calibration Method : CP-WK-E01,CP-WK-E02

Standard Instrument

<u>Instrument</u>	<u>Serial No.</u>	<u>Certificate No.</u>	<u>Due Date</u>	<u>Traceability to</u>
Multifunction Calibrator	752021	6011-KL-L1030-24	19-Sep-26	CMI

CMI : Czech Metrology Institute

This result calibrate was found accurate as shown on date place of calibrate only
This certificate is traceability to the International System of Unit (SI)

The reported expanded uncertainty it was multipiled by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Mr. Kritsada Oupparattha

Approved by :

Mr. Ratchadawut Rungravee
Authorized Signatory

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Calibration Results

Certificate No.: WK2504-402-60

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Calibration Results

DC Voltage Calibration

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
60 V	10 V	10.00 V	0.010 V	9.87 ~ 10.13 V
	30 V	30.00 V	0.010 V	29.67 ~ 30.33 V
	50 V	50.00 V	0.010 V	49.47 ~ 50.53 V
600 V	100 V	99.7 V	0.10 V	98.5 ~ 101.5 V
	300 V	299.4 V	0.20 V	296.1 ~ 303.9 V
	500 V	499.6 V	0.30 V	493.7 ~ 506.3 V

AC Voltage Calibration @ Test Frequency : 50 Hz

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
60 V	10 V	9.99 V	0.010 V	9.81 ~ 10.19 V
	30 V	29.94 V	0.010 V	29.51 ~ 30.49 V
	50 V	49.91 V	0.010 V	49.21 ~ 50.79 V
600 V	100 V	99.9 V	0.10 V	98.8 ~ 101.2 V
	300 V	300.0 V	0.20 V	296.8 ~ 303.2 V
	500 V	500.6 V	0.30 V	494.8 ~ 505.2 V

Resistance Calibration

(The zero offset calibration results were compensated on each ranges)

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
600 Ω	100 Ω	100.4 Ω	0.10 Ω	99.5 ~ 100.5 Ω
	300 Ω	300.4 Ω	0.10 Ω	299.5 ~ 300.5 Ω
	500 Ω	500.4 Ω	0.10 Ω	499.5 ~ 500.5 Ω
6 k Ω	1 k Ω	1.003 k Ω	0.0010 k Ω	0.977 ~ 1.023 k Ω
	3 k Ω	3.006 k Ω	0.0010 k Ω	2.937 ~ 3.063 k Ω
	5 k Ω	5.011 k Ω	0.0010 k Ω	4.897 ~ 5.103 k Ω
60 k Ω	10 k Ω	9.99 k Ω	0.010 k Ω	9.77 ~ 10.23 k Ω
	30 k Ω	29.95 k Ω	0.010 k Ω	29.37 ~ 30.63 k Ω
	50 k Ω	49.93 k Ω	0.010 k Ω	48.97 ~ 51.03 k Ω
600 k Ω	100 k Ω	100.0 k Ω	0.10 k Ω	97.7 ~ 102.3 k Ω
	300 k Ω	299.8 k Ω	0.10 k Ω	293.7 ~ 306.3 k Ω
	500 k Ω	499.5 k Ω	0.10 k Ω	489.7 ~ 510.3 k Ω

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Calibration Results

Certificate No.: WK2504-402-60

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Calibration Results

AC Current Calibration @ Test Frequency : 50 Hz

(Calibrated with 50 Turns Coil : > 20 A)

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
60 A	10 A	10.00 A	0.020 A	9.75 ~ 10.25 A
	20 A	20.00 A	0.040 A	19.55 ~ 20.45 A
	50 A	49.99 A	0.69 A	48.95 ~ 51.05 A
600 A	100 A	100.0 A	1.1 A	97.5 ~ 102.5 A
	300 A	300.3 A	3.5 A	293.5 ~ 306.5 A
	500 A	500.3 A	4.1 A	489.5 ~ 510.5 A

(X) Without Adjustment () After Adjustment

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**** End of Certificate****



Certificate of Calibration

Certificate No.: WK2411-404-87

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Customer : ZIRCOZ CO., LTD.

29/1293 Mooban Wararak Rangsit Khlongsam,
Moo.2, Khlong Sam, Khlong Luang, Pathumthani 12120

Instrument	: Thermometer with Sensor	Ambient Temperature	: (23 ± 2) °C
Manufacturer	: N/A	Humidity	: (50 ± 15) %RH
Model	: PCE-T390	Received Date	: 20-Nov-24
Serial No.	: H.463820	Calibrated Date	: 21-Nov-24
Identity No.	: ZTL-23-003	Issued Date	: 21-Nov-24
Range	: 30 °C to 200 °C	Calibrated Location	: In Lab
Resolution	: 0.1 °C		

Calibration Method : CP-WK-T09

Standard Instrument

<u>Instrument</u>	<u>Serial No.</u>	<u>Certificate No.</u>	<u>Due Date</u>	<u>Traceability to</u>
Platinum Resistance Thermometer	4507	PSL-T 0401/67	12-Feb-26	TISTR

TISTR : Thailand Institute of Scientific and Technological Research

This result calibrate was found accurate as shown on date place of calibrate only

This certificate is traceability to the International System of Unit (SI)

The reported expanded uncertainty it was multiplied by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Ms. Usa Phuangphiphat

Approved by :

Mr. Ratchadawut Rungravee

Authorized Signatory

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Calibration Results

Certificate No. : WK2411-404-87

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Calibration Result of the Accuracy

Function : Temperature Measurement with Sensor Type K T1

Range : 30 °C to 200 °C

Resolution : 0.1 °C

Unit : °C

Standard Reading	UUC Reading	Error	Uncertainty (± °C)
29.99	29.5	-0.49	0.18
100.02	99.5	-0.52	0.18
149.98	149.3	-0.68	0.18
199.99	199.2	-0.79	0.18

Function : Temperature Measurement with Sensor Type K T2

Range : 30 °C to 200 °C

Resolution : 0.1 °C

Unit : °C

Standard Reading	UUC Reading	Error	Uncertainty (± °C)
29.99	29.6	-0.39	0.18
100.02	99.3	-0.72	0.18
149.98	149.5	-0.48	0.18
199.99	199.3	-0.69	0.18

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Certificate of Calibration

Certificate No.: WK2411-404-86

Page 1 of 3

Customer : ZIRCOZ CO., LTD.
29/1293 Mooban Wararak Rangsit Khlongsam,
Moo.2, Khlong Sam, Khlong Luang, Pathumthani 12120

Instrument	: Thermometer with Sensor	Ambient Temperature	: (23 ± 2) °C
Manufacturer	: N/A	Humidity	: (50 ± 15) %RH
Model	: PCE-T390	Received Date	: 20-Nov-24
Serial No.	: H.451952	Calibrated Date	: 21-Nov-24
Identity No.	: ZTL-23-004	Issued Date	: 21-Nov-24
Range	: 30 °C to 200 °C	Calibrated Location	: In Lab
Resolution	: 0.1 °C		
Calibration Method	: CP-WK-T09		

Standard Instrument

<u>Instrument</u>	<u>Serial No.</u>	<u>Certificate No.</u>	<u>Due Date</u>	<u>Traceability to</u>
Platinum Resistance Thermometer	4507	PSL-T 0401/67	12-Feb-26	TISTR

TISTR : Thailand Institute of Scientific and Technological Research

This result calibrate was found accurate as shown on date place of calibrate only

This certificate is traceability to the International System of Unit (SI)

The reported expanded uncertainty it was multipiled by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Ms. Usa Phuangphiphat

Approved by :

Mr. Ratchadawut Rungravee

Authorized Signatory

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Calibration Results

Certificate No. : WK2411-404-86 Page 2 of 3

Calibration Result of the Accuracy

Function : Temperature Measurement with Sensor Type K T1
Range : 30 °C to 200 °C
Resolution : 0.1 °C

Unit : °C			
Standard Reading	UUC Reading	Error	Uncertainty (± °C)
29.99	30.1	0.11	0.18
99.99	99.9	-0.09	0.18
150.01	149.9	-0.11	0.18
199.98	199.9	-0.08	0.18

Function : Temperature Measurement with Sensor Type K T2
Range : 30 °C to 200 °C
Resolution : 0.1 °C

Unit : °C			
Standard Reading	UUC Reading	Error	Uncertainty (± °C)
29.99	30.1	0.11	0.18
99.99	100.1	0.11	0.18
150.01	150.1	0.09	0.18
199.98	200.1	0.12	0.18

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Calibration Results

Certificate No. : WK2411-404-86

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Calibration Result of the Accuracy

Function : Temperature Measurement with Sensor Type K T3

Range : 30 °C to 200 °C

Resolution : 0.1 °C

Unit : °C

Standard Reading	UUC Reading	Error	Uncertainty (± °C)
29.99	29.9	-0.09	0.18
99.99	99.9	-0.09	0.18
150.01	150.1	0.09	0.18
199.98	199.8	-0.18	0.18

Function : Temperature Measurement with Sensor Type K T4

Range : 30 °C to 200 °C

Resolution : 0.1 °C

Unit : °C

Standard Reading	UUC Reading	Error	Uncertainty (± °C)
29.99	29.9	-0.09	0.18
99.99	99.9	-0.09	0.18
150.01	149.9	-0.11	0.18
199.98	200.1	0.12	0.18

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**** End of Certificate****



Certificate of Calibration

Certificate No.: WK2411-404-90

Page 1 of 2

Customer : ZIRCOZ CO., LTD.
29/1293 Mooban Wararak Rangsit Khlongsam,
Moo.2, Khlong Sam, Khlong Luang, Pathumthani 12120

Instrument	: Digital Thermo-hygro Meter	Ambient Temperature	: (23 ± 2) °C
Manufacturer	: Mainscale	Humidity	: (50 ± 15) %RH
Model	: HTC-2	Received Date	: 20-Nov-24
Serial No.	: HTC2-2402024	Calibrated Date	: 20-Nov-24
Identity No.	: N/A	Issued Date	: 22-Nov-24
Range	: See to data	Calibrated Location	: In Lab
Resolution	: See to data		
Calibration Method	: CP-WK-T01		

Standard Instrument

Instrument	Serial No.	Certificate No.	Due Date	Traceability to
Temperature & Humidity Indicator with Sensor	HTD072K230577	CC287923000003451F	9-Jul-25	SANSEL

SANSEL CALIBRATION LABORATORIES

This result calibrate was found accurate as shown on date place of calibrate only

This certificate is traceability to the International System of Unit (SI)

The reported expanded uncertainty it was multipiled by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Ms. Usa Phuanghiphat

Approved by :

Mr. Ratchadawut Rungravee
Authorized Signatory

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Calibration Results

Certificate No. : WK2411-404-90

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Calibration Result of the Accuracy

Function : Temperature Measurement #IN

Range : 20 °C to 40 °C

Resolution : 0.1 °C

Unit : °C

STD Reading	UUC Reading	Error	Uncertainty (± °C)
19.99	19.9	-0.09	0.36
29.99	29.9	-0.09	0.36
40.02	40.1	0.08	0.36

Function : Temperature Measurement #OUT

Range : 20 °C to 40 °C

Resolution : 0.1 °C

Unit : °C

STD Reading	UUC Reading	Error	Uncertainty (± °C)
20.01	20.1	0.09	0.36
30.02	30.1	0.08	0.36
39.98	39.8	-0.18	0.36

Function : Humidity Measurement

Range : 40 %RH to 80 %RH

Resolution : 1 %RH

Unit : %RH

STD Reading	UUC Reading	Error	Uncertainty (± %RH)
40.05	42	1.95	1.3
60.02	58	-2.02	1.3
80.07	79	-1.07	1.3

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**** End of Certificate****



Certificate of Calibration

Certificate No.: WK2411-404-85

Page 1 of 2

Customer : ZIRCOZ CO., LTD.
29/1293 Mooban Wararak Rangsit Khlongsam,
Moo.2, Khlong Sam, Khlong Luang, Pathumthani 12120

Instrument	: Thermal Imager	Ambient Temperature	: (23 ± 2) °C
Manufacturer	: UNI-T	Humidity	: (50 ± 15) %RH
Model	: UTi712S	Received Date	: 20-Nov-24
Serial No.	: ATi0523120025	Calibrated Date	: 21-Nov-24
Identity No.	: ZTL-23-001	Issued Date	: 21-Nov-24
Range	: 30 °C to 200 °C	Calibrated Location	: In Lab
Resolution	: 0.1 °C		
Calibration Method	: CP-WK-T16		

Standard Instrument

<u>Instrument</u>	<u>Serial No.</u>	<u>Certificate No.</u>	<u>Due Date</u>	<u>Traceability to</u>
Platinum Resistance Thermometer	4507	PSL-T 0401/67	12-Feb-26	TISTR
Blackbody Calibration Source	201101008	TR-0095-23	19-Dec-25	NIMT

NIMT : National Institute of Metrology, Thailand

TISTR : Thailand Institute of Scientific and Technological Research

This result calibrate was found accurate as shown on date place of calibrate only

This certificate is traceability to the International System of Unit (SI)

The reported expanded uncertainty it was multipiled by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Ms. Usa Phuanghiphat

Approved by :

Mr. Ratchadawut Rungravee
Authorized Signatory

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Calibration Results

Certificate No. : WK2411-404-85

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Calibration Result of the Accuracy

Function : Temperature Measurement (Emissivity Setting : 0.95)

Range : 30 °C to 200 °C

Resolution : 0.1 °C

Unit : °C

Standard Value	UUC Reading	Error	Uncertainty (± °C)
30.0	30.3	0.3	0.20
100.0	100.5	0.5	0.20
200.0	200.4	0.4	1.1

(X) Without Adjustment () After Adjustment

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**** End of Certificate****



Certificate of Calibration

Certificate No.: WK2411-401-110

Page 1 of 2

Customer : ZIRCOZ CO., LTD.

29/1293 Mooban Wararak Rangsit Khlongsam,
Moo2., Khlong Sam, Khlong Luang Pathumthani 12120

Instrument	: Digital Torque Wrench	Ambient Temperature	: (23 ± 2) °C
Manufacturer	: ADEMA	Humidity	: (50 ± 15) %RH
Model	: 3-100 N.m	Received Date	: 20-Nov-24
Serial No.	: 158198	Calibrated Date	: 28-Nov-24
Equipment No.	: ZTL-23-002	Issued Date	: 28-Nov-24
Range	: (3-100) N.m	Calibrated Location	: In Lab
Resolution	: 0.01 N.m		
Calibration Method	: CP-WK-M07		

Standard Instrument

<u>Instrument</u>	<u>Serial No.</u>	<u>Certificate No.</u>	<u>Due Date</u>	<u>Traceability to</u>
Digital Torque Tester	111780	WK2403-301-101	19-Mar-25	WK Electric Co., Ltd.

This result calibrate was found accurate as shown on date place of calibrate only
This certificate is traceability to the International System of Unit (SI)

The reported expanded uncertainty it was multiplied by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Mr. Watchara Thongsorn

Approved by :

Mr. Ratchadawut Rungravee

Authorized Signatory

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Calibration Results

Certificate No. : WK2411-401-110

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Calibration Result of the Accuracy

Function : Torque Measurement

Range : (3-100) N m

Resolution : 0.01 N m

Unit : N m

Direction	UUC Setting	STD Reading	Error	Uncertainty (± N m)
CW	20.00	19.77	0.23	0.46
	60.00	59.34	0.66	0.46
	100.00	99.17	0.83	0.46
CCW	20.00	19.80	0.20	0.46
	60.00	59.23	0.77	0.46
	100.00	99.05	0.95	0.46

(X) Without Adjustment () After Adjustment

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**** End of Certificate ****



WK Electric Co., Ltd.

68/242 Moo 5, Sawaipracharaj Rd., Tumbol Ladsawai, Amphur Lamlucca, Pathumthani 12150

Tel. +66 2993 4773, +66 2153 7132-3 Fax. +66 2994 5509 E-mail : wk.calibrations@gmail.com www.wk-etc.com



Certificate of Calibration

Certificate No.: WK2406-307-2

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Customer : ZIRCOZ CO.,LTD.
29/1293 Mooban Wararak Rangsit Khlongsam, Moo.2, Khlong sam,
Khlong Luang, Pathumthani 12120

Instrument	: Digital Clamp Meter	Ambient Temperature	: (23 ± 2) °C
Manufacturer	: Kyoritsu	Humidity	: (50 ± 15) %RH
Model	: KEW 2117R	Received Date	: 2-Apr-24
Serial No.	: 1503170	Calibrated Date	: 2-Apr-24
Identity No.	: N/A	Issued Date	: 2-Apr-24
Range	: See to Data	Calibrated Location	: In Lab
Resolution	: See to Data		

Calibration Method : CP-WK-E01, CP-WK-E02

Standard Instrument

<u>Instrument</u>	<u>Serial No.</u>	<u>Certificate No.</u>	<u>Due Date</u>	<u>Traceability to</u>
Multi-Product Calibrator	5081802	EE-0016-23	6-Feb-25	NIMT

NIMT : National Institute of Metrology (Thailand).

This result calibrate was found accurate as shown on date place of calibrate only

This certificate is traceability to the International System of Unit (SI)

The reported expanded uncertainty it was multiplied by a coverage factor $k = 2$ providing a level of confidence approximately 95 %

Calibrated by : Mr. Kriksada Oupparattha

Approved by :

Mr. Ratchadawut Rungravee

Authorized Signatory

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Calibration Results

Certificate No.: WK2406-307-2

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Calibration Results

DC Voltage Calibration

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
60 V	10 V	10.00 V	0.010 V	9.87 ~ 10.13 V
	30 V	30.00 V	0.010 V	29.67 ~ 30.33 V
	50 V	50.00 V	0.010 V	49.47 ~ 50.53 V
600 V	100 V	99.7 V	0.10 V	98.5 ~ 101.5 V
	300 V	299.5 V	0.20 V	296.1 ~ 303.9 V
	500 V	499.7 V	0.30 V	493.7 ~ 506.3 V

AC Voltage Calibration @ Test Frequency : 50 Hz

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
60 V	10 V	9.98 V	0.010 V	9.81 ~ 10.19 V
	30 V	29.94 V	0.010 V	29.51 ~ 30.49 V
	50 V	49.89 V	0.010 V	49.21 ~ 50.79 V
600 V	100 V	99.9 V	0.10 V	98.8 ~ 101.2 V
	300 V	299.8 V	0.20 V	296.8 ~ 303.2 V
	500 V	500.4 V	0.30 V	494.8 ~ 505.2 V

Resistance Calibration

(The zero offset calibration results were compensated on each ranges)

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
600 Ω	100 Ω	100.5 Ω	0.10 Ω	99.5 ~ 100.5 Ω
	300 Ω	300.5 Ω	0.10 Ω	299.5 ~ 300.5 Ω
	500 Ω	500.5 Ω	0.10 Ω	499.5 ~ 500.5 Ω
6 k Ω	1 k Ω	1.003 k Ω	0.0010 k Ω	0.977 ~ 1.023 k Ω
	3 k Ω	3.007 k Ω	0.0010 k Ω	2.937 ~ 3.063 k Ω
	5 k Ω	5.010 k Ω	0.0010 k Ω	4.897 ~ 5.103 k Ω
60 k Ω	10 k Ω	9.99 k Ω	0.010 k Ω	9.77 ~ 10.23 k Ω
	30 k Ω	29.97 k Ω	0.010 k Ω	29.37 ~ 30.63 k Ω
	50 k Ω	49.95 k Ω	0.010 k Ω	48.97 ~ 51.03 k Ω
600 k Ω	100 k Ω	99.9 k Ω	0.10 k Ω	97.7 ~ 102.3 k Ω
	300 k Ω	299.8 k Ω	0.10 k Ω	293.7 ~ 306.3 k Ω
	500 k Ω	499.7 k Ω	0.10 k Ω	489.7 ~ 510.3 k Ω

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Calibration Results

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Calibration Results

AC Current Calibration @ Test Frequency : 50 Hz

(Calibrated with 50 Turns Coil : > 20 A)

UUC Range	Standard Value	UUC Reading	Uncertainty ($\pm$)	Tolerance Limit Values
60 A	10 A	10.04 A	0.020 A	9.75 ~ 10.25 A
	20 A	20.02 A	0.040 A	19.55 ~ 20.45 A
	50 A	50.04 A	0.69 A	48.95 ~ 51.05 A
	100 A	100.1 A	1.1 A	97.5 ~ 102.5 A
	300 A	300.3 A	3.5 A	293.5 ~ 306.5 A
600 A	500 A	500.5 A	4.1 A	489.5 ~ 510.5 A

(X) Without Adjustment () After Adjustment

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***** End of Certificate*****